

Service Operations Analytics

From incident audit trails to reliable reporting and demand planning.

6.11%
NYC test WAPE

97.6%
coverage of nominal 90% band

3 interactive views
incident, demand, resolution

Problem and business decision

A backlog total alone cannot explain workload changes, missing ownership, or future demand. The project supports queue investigation and capacity scenarios while exposing data quality and forecast uncertainty.

Data and engineering

A fixed 2016 UCI incident audit log is replayed with simulated duplicates, corrections and late deliveries. Current NYC 311 data supplies daily aggregate demand. Checks gate publication; immutable batches, versioned Parquet facts and dimensions, Spark SQL and Athena support reconciliation. S3 and CloudFront serve public aggregates.

Models and operating evidence

Time-ordered selection, calibration and a 42-day test compare weekday baselines with seasonal regression. Data cutoff: 2026-09-23; results above are a dated snapshot. The selected weekday baseline trails the four-week mean on this test (4.97% WAPE); earlier validation determines selection. A weather challenger remains separate. Published forecasts are retained for later scoring; retrospective charts are labeled separately. A censor-aware resolution model improves test probability error, but does not establish better resolution-time error.

What a visitor can use

Filter boroughs, compare uncertainty and error by weekday/lead, and move a capacity slider to see the complete demand-gap trade-off. Inspect backlog inflows/outflows and critical-versus-moderate survival curves. Download aggregate data and evaluations; no prediction endpoint or account is required.

Cost and limits

AWS reports \$0.38 estimated account-wide spend for September 1-25; this is not a project-only or full-month bill. Planning allowance: \$2-5/month at modest traffic, not a cap. No measured staffing savings or adoption claim. This is daily batch forecasting, not streaming; revised source records, small subgroups and dependent errors limit conclusions. The 44 MB incident workload does not demonstrate distributed scale.

Live: d17q1whdf7htno.cloudfront.net/forecast/

Code and definitions: github.com/RyanSingh0/service-operations-analytics-portfolio

Sources: UCI DOI 10.24432/C57S4H (CC BY 4.0); NYC Open Data 311 (erm2-nwe9).